Routing Is Least Learnable Where It Is Most Valuable: Bounds on Representation Routing for Web Agents

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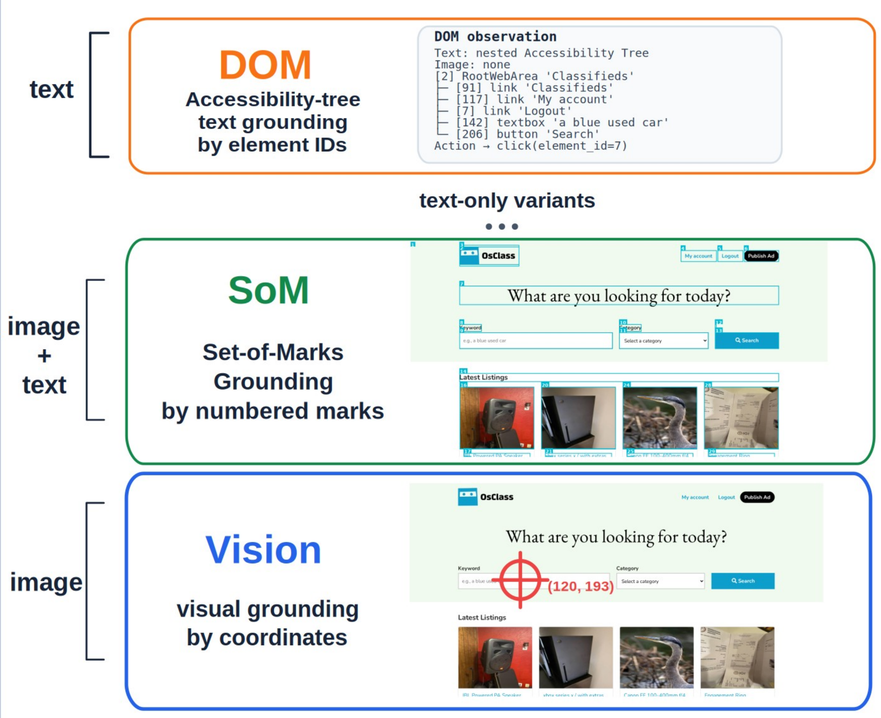
Six observation modes are complementary - but rerun noise and scarce success labels cap routing gains.

WHAT AN AGENT SEES, AND WHO CHOOSES

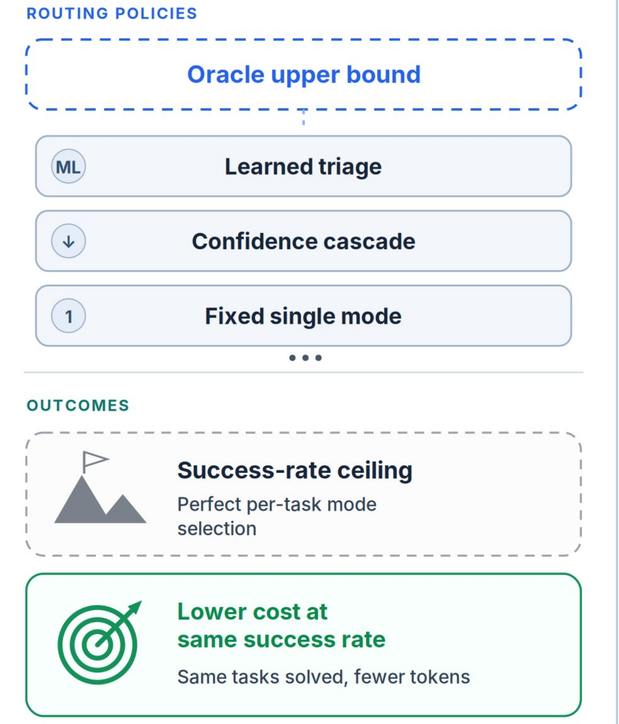
1 Agent ↔ Web Page



2 Grounding representations



3 Routing and outcomes

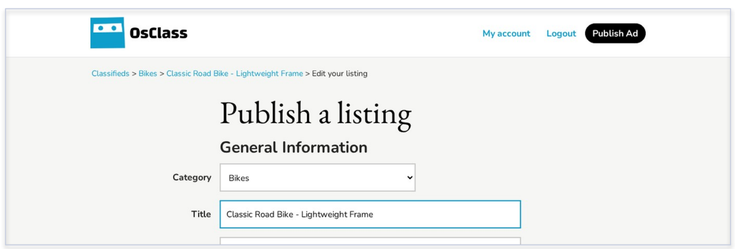


THE SAME TASK, SEEN TWO WAYS

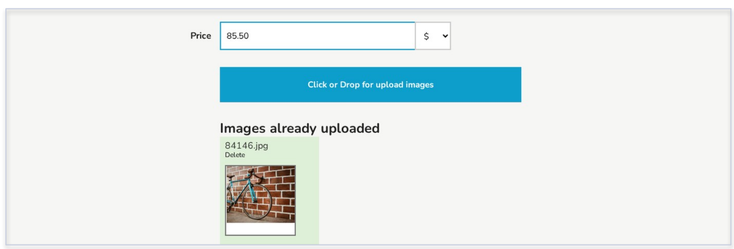
"Go to my listing of the blue bike and change the price to $85.50 — and say so in the description."

READ
page text only

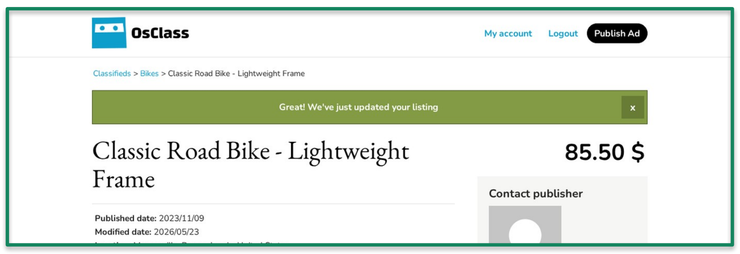
12 steps
8 pages seen
$0.09
solved



step 4 · type →
$0.020 spent



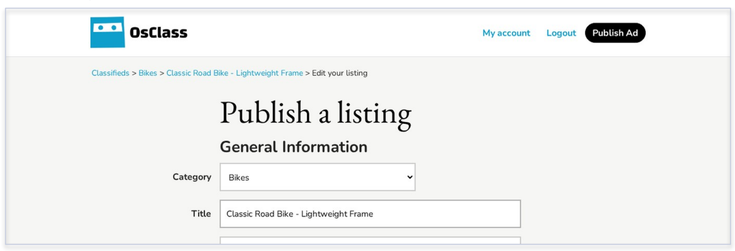
step 8 · type →
$0.050 spent



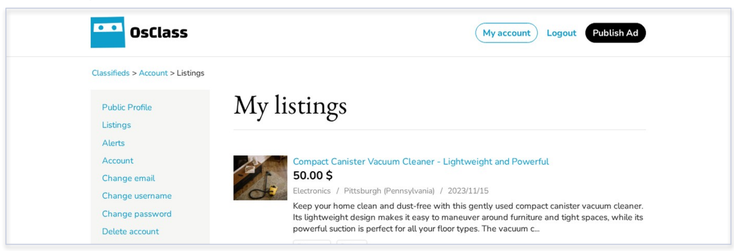
step 10 · click →
$0.086 spent

LOOK
screenshot only

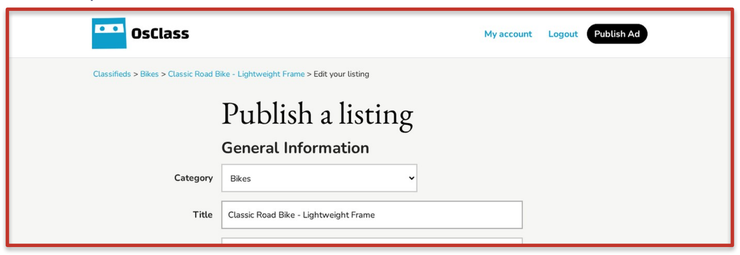
26 steps
9 pages seen
$0.10
gave up



step 2 · click →
$0.011 spent



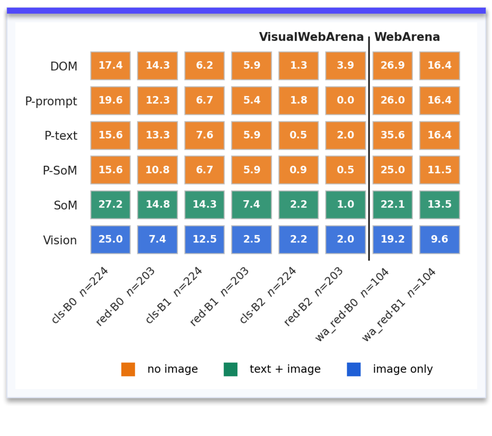
step 17 · click →
$0.071 · **left the form**



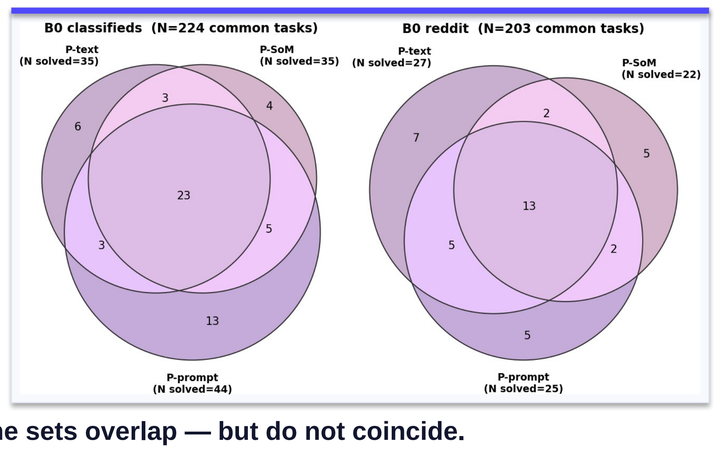
step 19 · click →
$0.079 · **back on frame 1**

Same start. READ finishes; LOOK loops back. Both outcomes repeat on an independent rerun.

1 · NO VIEW WINS EVERYWHERE

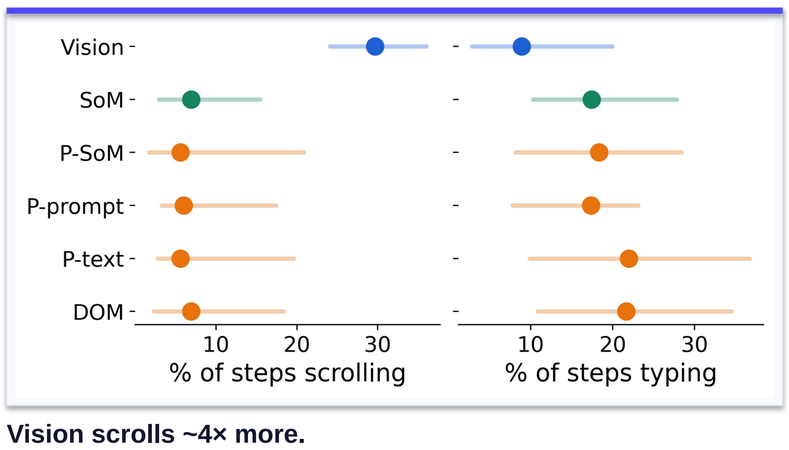


2 · THEY SOLVE DIFFERENT TASKS



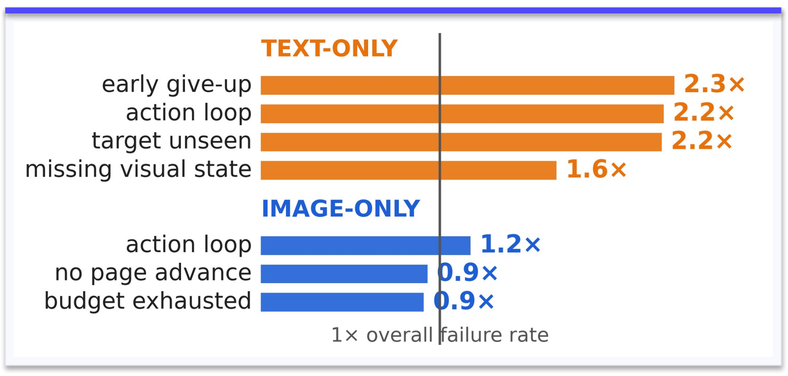
The sets overlap — but do not coincide.

3 · THEY BEHAVE DIFFERENTLY



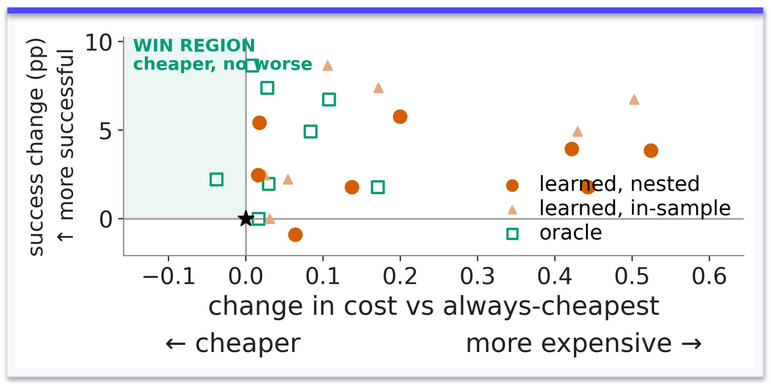
Vision scrolls ~4× more.

4 · AND THEY FAIL DIFFERENTLY



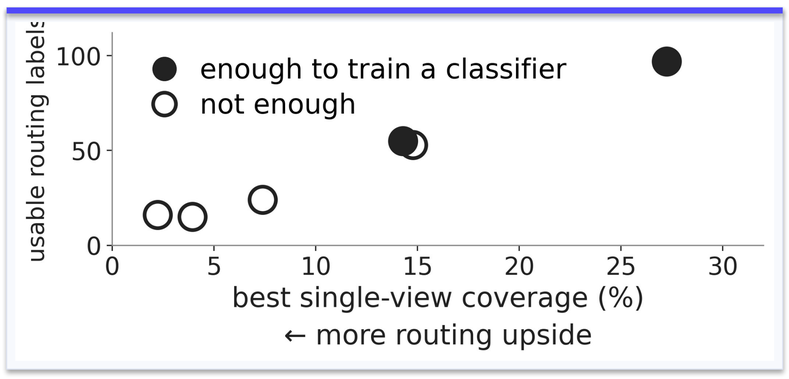
Text-only skews toward early give-up; image-only toward stalled progress.

5 · SO BUILD A ROUTER? NOT SO FAST.



Learned routers buy success only by spending more; only the hindsight oracle reaches the win region.

6 · AND THIS IS WHY



More routing upside, less usable training signal.

Accepted-paper correction: reruns flip 5-14% of outcomes, so oracle ceilings are partly noise.
Robust result: 9.5-30.6% lower cost at unchanged success; label supply and routing opportunity co-rise (r = 0.95).

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Paper, code & full results: github.com/Quarkgluonmixture/Cost-Aware-Routing-for-Web-Usage-Agents

REALM @ EMNLP 2026 · Non-archival poster · Paper 192
29 October 2026 · Budapest / online